

MULUND COLLEGE OF COMMERCE (AUTONOMOUS)

Programme: School of Business – Business
Administration

Class: SY Semester: III

Exam: External

Name of the Subject: AI POWERED MARKETING

Date of Exam: 04/10/2025

Marks: 30

Time: 1 hours

Instructions: 1) Figures to the right indicate maximum marks.

2) All Questions are Compulsory

Q1 **Read the following case study and answer the given questions** **10**

ShopEase, a mid-sized retail chain, is actively evaluating artificial intelligence (AI) solutions to strengthen its operations and enhance customer engagement. The leadership team faces challenges in managing stock levels, delivering personalized shopping experiences, setting competitive prices, and streamlining processes.

To structure their decision-making, ShopEase adopts an **AI Prioritization Matrix**, which categorizes solutions based on **implementation complexity** and **business impact**:

1. Low Complexity, Low Impact: Automated email responders for frequently asked customer queries. Easy to roll out, but limited influence on strategic outcomes.

2. Low Complexity, High Impact: AI-driven inventory alerts and sales prediction tools. Simple to integrate and can drastically reduce wastage and stock shortages.

3. High Complexity, Low Impact: AI-enabled smart shelving robots for restocking aisles. Expensive and technically demanding, but benefits are mostly invisible to customers.

4. High Complexity, High Impact: An advanced AI ecosystem that integrates customer behavior analytics, predictive supply chain management, automated pricing optimization, and logistics route planning. This system could revolutionize efficiency and profitability but requires significant time, money, and expertise.

A Which AI initiative should ShopEase pursue first if the goal is to achieve quick improvements without heavy investments? Why? **05**

B What risks and rewards could ShopEase face if it invests in a High Complexity, High Impact AI system? **05**

Q2 **Read the following case study and answer the given questions** **10**

StyleHub, a growing lifestyle and apparel retailer, is struggling to maintain its market position amidst rising competition and evolving consumer expectations. To strengthen its marketing effectiveness, the company is evaluating how Artificial Intelligence (AI) can be embedded into its strategy.

The leadership team has identified five core AI-driven marketing components:

1. Customer Insights (Data Analysis): Using AI to study customer demographics, purchase journeys, and online behavior to uncover actionable trends.

2. Marketing Automation: Implementing automated workflows for emails, social posts, and campaign bidding to reduce manual work.

3. Hyper-Personalization: Delivering individualized recommendations and offers aligned with each shopper's preferences and past activity.

4. Predictive Intelligence: Leveraging AI models to forecast future demand, seasonal trends, and consumer behavior shifts.

5. Adaptive Pricing (Dynamic Pricing): Adjusting prices in real time based on demand surges, competition, and inventory levels to capture maximum revenue.

A Which AI marketing component would have the most immediate effect on StyleHub's customer engagement, and why? **05**

B How can StyleHub leverage Predictive Intelligence and Adaptive Pricing to strengthen its profitability? **05**

Q 3 **Read the following case study and answer the given questions** **10**

ElectroMart, a nationwide consumer electronics retailer, is facing challenges in retaining loyal customers and attracting new ones in a highly competitive market. To improve its marketing effectiveness, the company is exploring the use of **AI algorithms** to strengthen customer segmentation, refine targeting, and deliver a superior shopping experience.

After discussions with its data science team, ElectroMart is considering the following AI approaches:

1. Logistic Regression: Estimates the probability of a purchase based on previous browsing patterns, purchase history, and demographic attributes.

2. Decision Trees: Segments customers into groups based on behaviors such as frequency of purchases, product categories browsed, and responsiveness to promotions.

3. K-Means Clustering: Groups customers into natural clusters (e.g., budget-conscious buyers, tech enthusiasts, brand-loyal customers) to support targeted campaigns.

4. Natural Language Processing (NLP): Examines online reviews, feedback forms, and social media mentions to uncover customer sentiment and evolving preferences.

A How can ElectroMart use K-Means Clustering and Decision Trees to improve customer targeting? **05**

B How can ElectroMart leverage NLP and Logistic Regression to enhance customer experience and predict purchase behavior? **05**